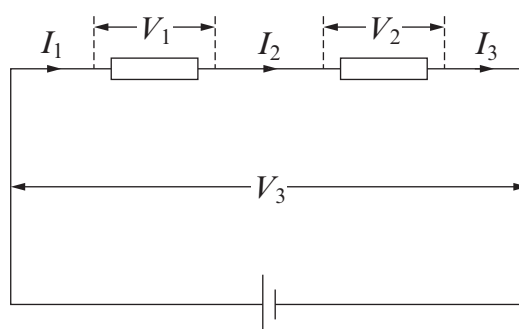


2.



(a) For the series circuit shown, write equations that give the relationships between:

(i) the currents (only);

[1]

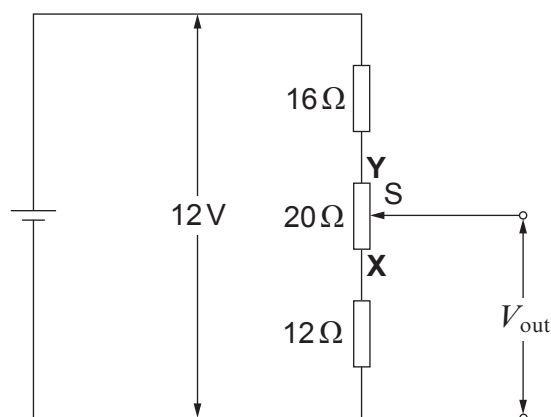
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(ii) the potential differences (only).

[1]

.....

(b) The diagram shows a circuit that can produce a range of output pds, V_{out} . The middle resistor is a uniform carbon conductor, XY, of resistance $20\ \Omega$. A sliding contact, S, can be moved up and down, making contact with the conductor at any point along it.



(i) Show clearly that when the sliding contact is at Y, $V_{\text{out}} = 8.0\text{V}$.

[2]

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(ii) Determine the value of V_{out} when the sliding contact is at X.

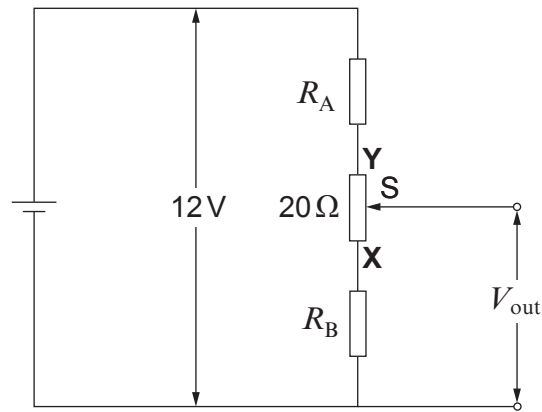
[1]

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- (c) The circuit in (b) is to be modified to produce V_{out} ranging from 2V to 10V. The 12V battery and the 20Ω resistor with sliding contact are still to be used.



Design the new circuit, by giving the values for R_A and R_B .

[2]

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